

Physics-guided search and graph learning for conductive- particle allocation in a polydisperse packed bed

Ali Khalifa and

Technical University of Munich, Freising, Germany.

Abstract

A packed thermal store containing a limited amount of highly conductive material can perform differently depending on where that material is placed. This study isolates that placement effect while keeping packing geometry, porosity, particle-size distribution and conductive-material volume fixed. Ten independent 500-particle packings with three diameter classes (1.5, 2.0 and 2.5 mm) were generated by the discrete-element method at a nominal porosity of 0.38. A verified steady contact-network solver evaluated random placement, contact-degree and baseline heat-path rankings, and a fixed-budget same-size swap search. The best searched assignments increased effective conductivity by $56.3 \pm 4.1\%$ relative to random placement. A graph neural network (GNN) was then trained to imitate the recurring selections of five independent search restarts and was tested by leave-one-packing-out cross-validation. A non-graph multilayer perceptron (MLP) received exactly the same particle features. On unseen packings, the MLP and GNN obtained $k_{\text{eff}} = 0.13137 \pm 0.00583$ and $0.14189 \pm 0.00731 \text{ W m}^{-1} \text{ K}^{-1}$, respectively, compared with 0.12208 ± 0.00467 for the transparent heat-path rule. The GNN exceeded both baselines in all ten held-out packings and recovered $50.7 \pm 7.7\%$ of the gain between random placement and the best searched assignment; the MLP recovered $33.6 \pm 4.3\%$. Once trained, the GNN required one homogeneous feature solve, approximately 4 ms of CPU inference and one verification solve, whereas teacher generation used about 14829 distinct thermal evaluations per packing. Mechanism analysis showed that the improvement arises from cooperative, vertically useful conductive paths rather than local coordination alone. The results establish graph learning as a useful fast allocation model for unseen realizations of the investigated stationary, contact-dominated bed. Fluid convection, radiation and transfer to other packing specifications remain outside the present scope.

Keywords: packed-bed thermal energy storage; granular heat conduction; discrete-element method; graph neural network; material allocation; inverse design

1 Introduction

Heat transfer through a packed bed depends on particle conductivity, particle size, porosity, contact area and the organization of the contact network. Classical theory and thermal discrete-element models describe how these factors determine heat flow through a specified granular structure [6, 7, 8, 9]. Most such calculations are forward problems: the packing and material distribution are prescribed, and the resulting temperature field or effective conductivity is calculated.

Many engineering beds, however, contain a limited amount of an expensive or specialized conductive phase. In that situation, the practical question is not only how much conductive material is present, but where it should be placed. Random mixing can put conductive particles in regions that carry little net heat. Selecting particles with many contacts may also be inefficient, because a highly connected particle can lie on a side branch rather than on a route that links the hot and cold boundaries.

Inverse design on graph-structured materials has been considered for architected lattices [10]. A packed bed creates a different problem: the graph is not freely drawn or deformed, but is the mechanically admissible contact network produced by particle packing. The geometry remains fixed and only the particle material labels are changed. This gives a controlled question:

At fixed porosity, particle-size distribution and conductive-material volume, can the effective conductivity of a packed bed be increased substantially by placing conductive particles according to the global heat-transfer network?

We address this question using ten independent DEM packings, a verified steady thermal contact network and four increasingly informed allocation strategies. We then ask whether a learned model can transfer the allocation rule to a previously unseen packing without performing thousands of trial thermal solves. An edge-aware GNN is compared with a non-graph MLP supplied with exactly the same particle features. This comparison separates the value of contact-network message passing from the value of the input features themselves.

Figure 1 summarizes the physical problem. DEM first fixes the particle positions and contacts. The material budget is then allocated without moving any particle or changing the packing porosity and particle-size distribution. The present paper considers heat conduction through solid contacts only; stagnant-fluid conduction is reserved for a later, separately validated extension. The investigated millimetre-scale spheres represent the lower end of particle sizes used in solid-particle thermal storage and provide a controlled

model bed rather than a calibrated representation of one particular commercial storage material [2, 5].

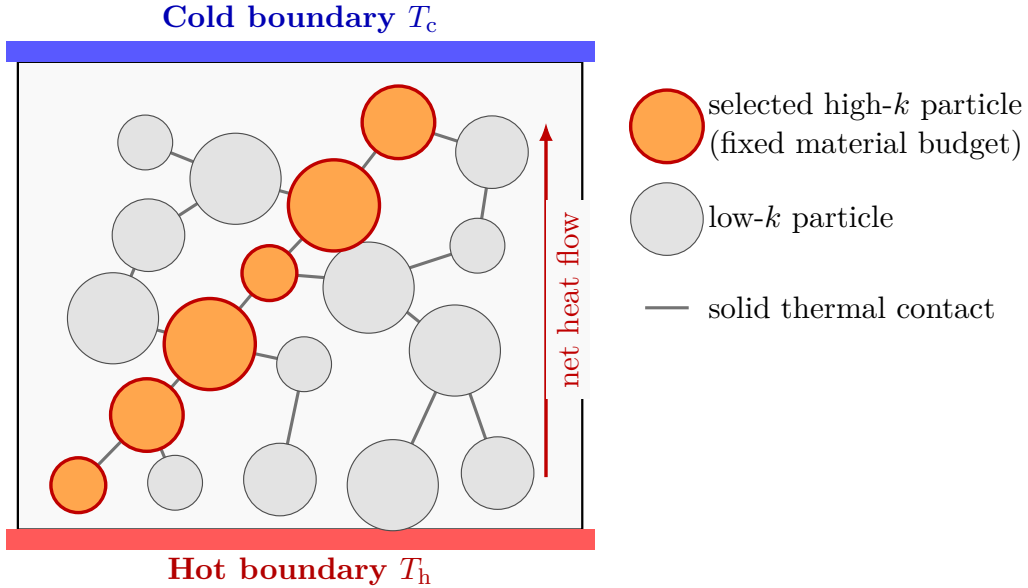


Figure 1: Schematic of the inverse allocation problem for a polydisperse packed bed. The three illustrated particle sizes represent the simulated diameter classes of 1.5, 2.0 and 2.5 mm. DEM fixes the particle positions, particle-size distribution and contact network. A prescribed number of particles from each size class receives the high-conductivity material, while the packing geometry and total conductive-material volume remain unchanged. Heat is conducted through the fixed solid-contact network.

2 Methods

2.1 Controlled DEM packing ensemble

The packings were generated with LAMMPS (22 July 2025, Update 5) using spherical particles and a non-cohesive Hertz–Mindlin contact law [1, 14]. Each realization contains 500 particles in a cubic cell of side $H = 15.3706$ mm. The x and y directions are periodic, whereas flat mechanical walls at $z = 0$ and $z = H$ confine the bed. Gravity was omitted because the purpose was to generate statistically different, equal-volume representative contact networks without a vertical compaction gradient.

The number-based particle-size distribution (PSD) is 100 particles of diameter 1.5 mm, 300 of diameter 2.0 mm and 100 of diameter 2.5 mm, corresponding to 20/60/20% by number. These three discrete sizes introduce controlled polydispersity while retaining exact size-class quotas during material allocation. Diameters of 1–100 mm occur across solid-particle storage concepts; the present 1.5–2.5 mm interval lies at the small-particle

end, where large surface area is attractive for heat exchange but pressure drop becomes an important system-level constraint [2, 3]. The selected density, 2500 kg m^{-3} , is representative of mineral and ceramic sensible- heat media. These choices define a generic model system: they support a mechanistic investigation of contact-network allocation but are not claimed to reproduce one named rock, slag or ceramic.

The exact undeformed solid volume, $V_s = \sum_i \pi d_i^3/6$, was used to set the cell volume through $H^3 = V_s/(1-\varepsilon)$ with target porosity $\varepsilon = 0.38$. This value is consistent with a dense random sphere packing and is relevant to stationary packed-bed stores, in which packing fraction and particle arrangement affect both thermal transport and thermomechanical loading [3, 4]. Because the particle counts, diameters and cell volume are exact, all realizations have the same nominal porosity and PSD by construction rather than by post-hoc matching.

Table 1 gives the mechanical and numerical parameters. The normal force is Hertzian, the history-dependent tangential force follows the Mindlin model with Coulomb limit, and Tsuji damping uses the specified restitution coefficient. The same law is used at the two vertical walls. The Young modulus of 50 MPa is deliberately softened relative to many rocks and ceramics to make quasistatic packing generation affordable. It is therefore a numerical packing parameter, not a calibrated storage-material property. This qualification matters because the subsequent thermal contact radius is computed from the final DEM overlap; sensitivity to stiffness is accordingly retained as a limitation.

Particle centres were initially sampled from the seed-dependent uniform random distribution while all diameters were reduced to 5% of their final values. The three classes were expanded linearly to their prescribed diameters over 300000 steps, followed by 200000 steps of damped relaxation at final size. Translational and rotational equations were advanced with the explicit velocity-Verlet scheme implemented by `fix nve/sphere` [15]. The time step $2.0 \times 10^{-7} \text{ s}$ is approximately 0.7% of the estimated Rayleigh time of the smallest sphere for the stated softened elastic properties, providing a conservative resolution of the contact dynamics. Neighbour lists were checked every step and loss of any particle was treated as an error.

Ten seeds generated ten different initial centre arrangements and hence ten different contact networks. Particle counts, sizes, material parameters, cell dimensions, growth schedule and relaxation schedule were otherwise identical. Final particle volumes and the 100/300/100 counts were written to a packing summary and checked against their requested values. The resulting graphs contained 1325–1382 particle–particle contacts, and every packing contained one large contact component connecting the lower and upper thermal walls.

Table 1: DEM packing-generation parameters used for every realization.

Quantity	Value or model
Particle count and diameters	500; 100 at 1.5 mm, 300 at 2.0 mm, 100 at 2.5 mm
Density	2500 kg m ⁻³
Target porosity and cell side	0.38; 15.3706 mm
Boundary conditions	periodic x, y ; fixed flat walls at $z = 0, H$; no gravity
Normal contact	Hertz material model; $E = 5.0 \times 10^7$ Pa, $\nu = 0.25$
Tangential contact	Mindlin history with rescaling; friction coefficient $\mu = 0.50$
Collision damping	Tsuji model; restitution coefficient $e = 0.30$; tangential damping multiplier 1.0
Time integration	explicit velocity-Verlet for translating and rotating spheres (nve/sphere)
Time step	2.0×10^{-7} s
Packing schedule	diameter growth: 300000 steps; final-size relaxation: 200000 steps
Viscous relaxation	10^{-3} during growth; 0.10 after growth (LAMMPS SI units)

2.2 Steady thermal contact network

Each particle is represented by a graph node and every DEM contact by a thermal edge. For an internal particle i , steady energy conservation is

$$\sum_{j \in \mathcal{N}_i} G_{ij}(T_j - T_i) = 0, \quad (1)$$

where G_{ij} is the contact conductance. For two particles with conductivities k_i and k_j and contact radius a_{ij} , the constriction model is

$$G_{ij} = \frac{2a_{ij}k_ik_j}{k_i + k_j}, \quad a_{ij} = \sqrt{R_{ij}^*\delta_{ij}}, \quad (2)$$

where $R_{ij}^* = R_iR_j/(R_i + R_j)$ and δ_{ij} is the DEM overlap. Fixed temperatures of 301 and 300 K were applied at the lower and upper walls. The effective conductivity follows from

$$k_{\text{eff}} = \frac{QH}{A(T_h - T_c)}, \quad (3)$$

where Q is the wall heat rate, H is bed height and A is cross-sectional area. Wall conductance uses the corresponding sphere-plane contact radius and an infinite wall conductivity, so the finite resistance lies on the particle side. The assembled balances

form a sparse symmetric conductance system. Components connected to neither thermal wall have no unique absolute temperature and carry no boundary-to-boundary heat; they are identified by graph connectivity, omitted from the linear solve and reported separately. The remaining sparse linear system is solved directly in double precision. Effective conductivity is calculated from both wall heat rates, whose relative mismatch provides the energy-balance error.

The present paper uses solid-contact conduction only. A preliminary stagnant-air gap model was tested, but its predicted conductivity depended too strongly on an arbitrary neighbour cutoff. Those results were therefore excluded from the allocation conclusions. This choice makes the present scope a stationary, contact-dominated bed rather than a flowing or convective system.

2.3 Fixed material budget

The low- and high-conductivity materials were assigned $k_L = 1$ and $k_H = 10 \text{ W m}^{-1} \text{ K}^{-1}$. A binary variable x_i indicates whether particle i receives the high-conductivity material:

$$k_i = k_L + x_i(k_H - k_L). \quad (4)$$

Because the bed is polydisperse, selecting the same total number of particles does not automatically select the same material volume. Every allocation was therefore constrained to contain 10 small, 30 medium and 10 large high-conductivity particles. The selected conductive-particle volume was $2.2514747 \times 10^{-7} \text{ m}^3$ in every calculation. This constraint preserves both the conductive-particle count and its size distribution.

2.4 Allocation strategies

Four strategies were compared on every fixed packing:

1. **Random placement:** 100 independent assignments satisfying the size-class quotas. Their mean and standard deviation define the undesigned reference.
2. **Degree ranking:** within each size class, particles with the most contacts are selected first.
3. **Heat-path ranking:** the homogeneous low-conductivity bed is solved first. Each particle receives the score

$$S_i = \sum_j |G_{ij}(T_i - T_j)|, \quad (5)$$

which measures the baseline heat passing through its incident contacts. The highest-scoring particles are selected within each size class.

4. **Fixed-budget same-size swap search:** the search starts from the better of degree and heat-path ranking. One selected and one unselected particle of the same size class exchange their material labels. The proposal is retained only when it increases k_{eff} . Each of five independent restarts is limited to exactly 3000 proposed swaps.

Here, “same-size swap” describes the search move, whereas “fixed budget” describes its stopping rule; together they define one algorithm, not two different methods. Restricting each exchange to one size class preserves the number, size distribution and total volume of high-conductivity particles. Similar exchange strategies are widely used in combinatorial optimization [11, 12], although our implementation is adapted specifically to the packed-bed thermal problem. Because only a fixed number of proposals is tested, the method identifies a good allocation but does not guarantee the global optimum. We report both the mean final result across the five restarts and the best final result among them. Performance is evaluated relative to the mean result obtained from random allocations in the same packing:

$$I = 100 \frac{k_{\text{eff}} - \bar{k}_{\text{eff,random}}}{\bar{k}_{\text{eff,random}}} [\%]. \quad (6)$$

2.5 Graph-learning allocation models

The fixed-budget same-size swap search provides high-quality assignments but requires repeated thermal solutions. We therefore tested whether a supervised model could learn a fast particle-ranking rule and transfer it to a packing not used during training. One graph was constructed for each DEM realization. Nodes are particles; each undirected DEM contact is stored as two directed edges so that messages can pass in both directions.

Inputs and target. Each node has eight features: diameter divided by the reference diameter, normalized vertical position, contact degree, hot-wall-contact flag, cold-wall-contact flag, membership in a thermally wall-anchored component, normalized temperature in the homogeneous k_L solution, and the normalized baseline heat-path score. Each directed edge has six features: normalized contact radius; the three normalized minimum-image components Δx , Δy and Δz ; absolute vertical alignment $|\Delta z|/d_{ij}$; and normalized absolute contact heat rate in the homogeneous solution. All inputs are available before the high-conductivity material is allocated.

The teaching signal is obtained from the five final same-class swap-search restarts for the corresponding packing. For particle i , the soft target $y_i \in \{0, 0.2, \dots, 1\}$ is the fraction

of restarts that selected it. This captures recurring choices instead of forcing the model to copy only one stochastic best assignment. Binary membership of the best restart is retained only for Jaccard evaluation and is not an input. The soft targets sum exactly to 10, 30 and 10 within the three particle classes.

Same-feature MLP benchmark. The MLP processes each particle independently and tests whether node features alone are sufficient. It receives the same eight node features as the GNN. It contains two fully connected hidden layers of width 64, with ReLU activation and dropout probability 0.10 after each layer, followed by one scalar ranking score. It receives no contact list or edge feature. Therefore, a consistent improvement of the GNN over the MLP measures the value of explicit contact-network message passing rather than access to additional particle inputs.

Edge-aware GNN. The GNN first maps the eight node features to a 64-dimensional hidden state. Three message-passing layers are then applied. In layer ℓ , the directed message from particle j to particle i is

$$\mathbf{m}_{ji}^{(\ell)} = \phi_m^{(\ell)}([\mathbf{h}_j^{(\ell)}, \mathbf{h}_i^{(\ell)}, \mathbf{e}_{ji}]), \quad (7)$$

where ϕ_m is a two-layer 64-unit ReLU network and $\mathbf{e}_{ji}$ contains the six edge features. Incoming messages are averaged, concatenated with the current node state and passed through a two-layer update network. A residual connection and layer normalization give

$$\mathbf{h}_i^{(\ell+1)} = \text{LayerNorm}[\mathbf{h}_i^{(\ell)} + \phi_u^{(\ell)}([\mathbf{h}_i^{(\ell)}, \text{mean}_{j \in \mathcal{N}_i} \mathbf{m}_{ji}^{(\ell)}])]. \quad (8)$$

A final 64-unit ReLU layer, dropout of 0.10 and a scalar output layer produce the particle score. The implementation uses PyTorch directly. Its relational structure follows the graph-network principle of learning from entities, relations and their interactions [13].

Training and constrained inference. Both models use the same training procedure. Features are standardized using means and standard deviations calculated from the training packings only. The loss is the mean of three type-wise listwise cross-entropies between predicted scores and normalized soft targets, plus 0.20 times a soft binary cross-entropy. Type-wise ranking is required because inference must satisfy the fixed 10/30/10 quota. AdamW is used with learning rate 10^{-3} , weight decay 10^{-4} and gradient-norm clipping at 5. Training is limited to 400 epochs and stops when validation loss has not improved by at least 10^{-6} for 60 epochs.

Evaluation is leave-one-packing-out, never a random split of particles. In each of ten folds, one complete packing is the unseen test graph, the preceding packing in the ordered seed list is the validation graph (with circular wrap), and the remaining eight graphs are used for training. Each model is trained three times per fold with deterministic but different seeds. Because logits from independent runs can have different offsets and scales, scores are converted to normalized ranks separately within each size class and averaged across the three repeats. The final allocation selects the 10, 30 and 10 highest ensemble ranks in the respective classes. Each predicted ID list is passed to the original thermal solver; no learned approximation of k_{eff} is used.

The primary ML endpoint is thermally evaluated k_{eff} on the held-out packing. The fraction of the searched improvement recovered by a model is

$$f_{\text{rec}} = \frac{k_{\text{eff,model}} - \bar{k}_{\text{eff,random}}}{k_{\text{eff,best}} - \bar{k}_{\text{eff,random}}}. \quad (9)$$

Performance relative to heat-path ranking and Jaccard overlap with the best searched ID set are also reported. Jaccard overlap is secondary because distinct particle sets can create similarly effective parallel heat paths.

Table 2: Machine-learning settings used in every held-out-packing fold.

Item	Setting
Data split	10 outer folds: 1 test graph, 1 validation graph and 8 training graphs
Repeated training	3 deterministic seeds per model and fold
MLP	8 inputs; hidden widths 64–64; ReLU; dropout 0.10; scalar score
GNN	8 node and 6 edge inputs; width 64; 3 mean-aggregation message layers; residual update and layer normalization
Target	particle selection frequency across five search restarts
Loss	type-wise listwise cross-entropy +0.20 soft BCE
Optimizer	AdamW; learning rate 10^{-3} ; weight decay 10^{-4} ; gradient clipping 5
Stopping	maximum 400 epochs; patience 60; validation tolerance 10^{-6}
Ensemble	mean normalized within-type rank from 3 repeats
Inference constraint	exactly 10/30/10 selected particles by size class
Final evaluation	original sparse thermal solver on the predicted ID list

2.6 Numerical verification and statistical reporting

The sparse solver was tested for energy conservation and linear scaling. On a representative packing, doubling all particle conductivities doubled both heat rate and k_{eff} . Increasing the imposed temperature difference by a factor of ten increased heat rate by ten while leaving k_{eff} unchanged. Temperatures remained between the boundary values.

The ten DEM realizations are treated as independent packing samples. Results are reported as mean $\pm$ sample standard deviation. For the improvement over random placement, a 95% confidence interval was calculated with the Student t distribution. Each best particle-ID assignment was also passed to an independent call of the thermal solver.

The complete computational workflow is shown in Fig. 2. DEM is required only once for each packing. The same contact graph can subsequently be evaluated for many material allocations, which makes controlled comparison and fixed-budget same-size swap search possible without regenerating the packing.

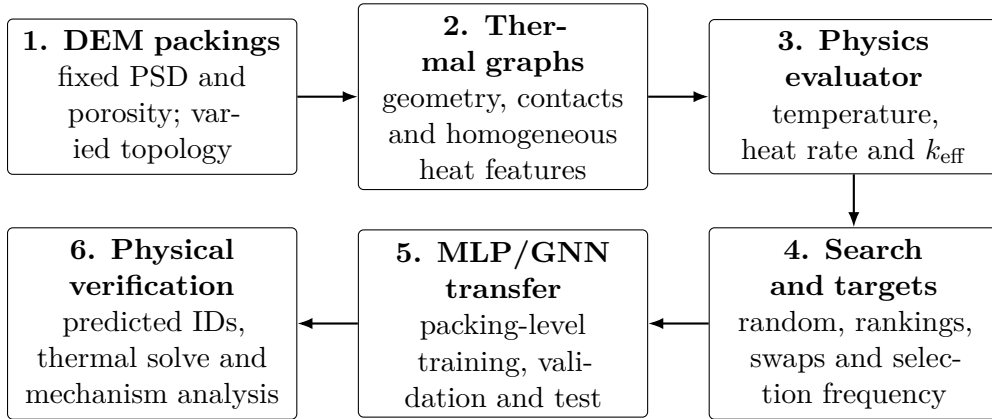


Figure 2: Computational workflow. DEM is run once per packing. The thermal network supplies physical baselines and search targets; MLP and GNN allocations are evaluated on complete unseen packings by the unchanged thermal solver.

3 Results

3.1 Homogeneous packing variability

With all particles assigned $k = 1 \text{ W m}^{-1} \text{ K}^{-1}$, the ten packings gave

$$k_{\text{eff,hom}} = 0.09329 \pm 0.00351 \text{ W m}^{-1} \text{ K}^{-1}. \quad (10)$$

The range was 0.08615–0.09717 $\text{W m}^{-1} \text{ K}^{-1}$ and the coefficient of variation was 3.77%. The mean coordination number was 5.393 ± 0.066 , with a smaller coefficient of variation

of 1.22%. Thus, packings with nearly identical contact counts still showed measurable thermal variation, indicating that contact arrangement and conductance weights matter.

Between one and eleven particles per packing were outside the wall-anchored contact component. They carry no heat in the contact-only model and were excluded from the linear system. Every packing nevertheless contained one large component spanning both thermal walls.

3.2 Allocation performance across ten packings

Table 3 shows the ensemble results. Adding 10% conductive particles at random raised the mean conductivity to $0.11046 \pm 0.00413 \text{ W m}^{-1} \text{ K}^{-1}$. Degree ranking produced only a small additional improvement and was slightly worse than the random mean in one packing. Heat-path ranking was better than random in all ten packings.

Table 3: Performance across ten independent DEM packings. Improvement is calculated relative to the controlled random mean in the same packing.

Allocation	$k_{\text{eff}} [\text{W}/(\text{m K})]$	Mean improvement
Random mean	0.11046 ± 0.00413	reference
Degree ranking	0.11310 ± 0.00425	$2.4 \pm 2.5\%$
Heat-path ranking	0.12208 ± 0.00467	$10.5 \pm 2.3\%$
Fixed-budget swap	0.16868 ± 0.00692	$52.7 \pm 3.5\%$
search, five-restart mean		
Fixed-budget swap	0.17263 ± 0.00787	$56.3 \pm 4.1\%$
search, best of five		

The best result from the five fixed-budget swap-search restarts improved on random placement in every realization. The smallest improvement was 50.8% and the largest was 60.9%. The mean improvement was 56.3%, with a 95% confidence interval of 53.4–59.2%. The corresponding improvement over the homogeneous low-conductivity bed was $85.1 \pm 4.9\%$.

The mean of the five final fixed-budget swap-search values was also substantially above random in every packing, with improvements between 48.7% and 58.1%. This shows that the conclusion does not depend only on selecting the best of five search trajectories.

3.3 Transfer to unseen packings with MLP and GNN

Table 5 reports the corrected three-repeat rank ensembles. Both learned models outperformed the transparent heat-path ranking in all ten held-out packings. The MLP reached $k_{\text{eff}} = 0.13137 \pm 0.00583 \text{ W m}^{-1} \text{ K}^{-1}$, corresponding to an improvement of $18.9 \pm 2.9\%$ over random placement and recovery of $33.6 \pm 4.3\%$ of the best-search gain. The edge-aware

Table 4: Per-packing controlled allocation results.

Seed	Random mean	Heat path	Best swap search	Gain [%]
18427	0.11123	0.12186	0.17895	60.9
29173	0.10194	0.11045	0.15600	53.0
37811	0.11317	0.12510	0.18102	60.0
46549	0.11216	0.12153	0.16918	50.8
55291	0.11330	0.12602	0.18219	60.8
64109	0.10562	0.12018	0.16895	60.0
72869	0.11075	0.12669	0.17540	58.4
81647	0.11472	0.12467	0.17498	52.5
90439	0.11402	0.12369	0.17334	52.0
99223	0.10774	0.12058	0.16628	54.3

GNN reached $0.14189 \pm 0.00731 \text{ W m}^{-1} \text{ K}^{-1}$, or $28.4 \pm 3.8\%$ above random, and recovered $50.7 \pm 7.7\%$ of the searched gain. Its recovery fraction ranged from 41.9% to 69.0%.

Table 5: Allocation performance on complete held-out packing graphs. Values are mean $\pm$ sample standard deviation over ten outer folds.

Method	k_{eff} [W/(m K)]	vs. random [%]	vs. heat path [%]	f_{rec} [%]
Heat-path ranking	0.12208 ± 0.00467	10.5 ± 2.3	reference	—
Same-feature MLP	0.13137 ± 0.00583	18.9 ± 2.9	7.6 ± 1.9	33.6 ± 4.3
Edge-aware GNN	0.14189 ± 0.00731	28.4 ± 3.8	16.2 ± 3.7	50.7 ± 7.7
Fixed-budget swap search, best of five	0.17263 ± 0.00787	56.3 ± 4.1	—	100

The GNN exceeded the MLP in every held-out packing and increased mean effective conductivity by 8.0% relative to the MLP. This is the central ablation result: because both models use identical node inputs, explicit contact and edge information provides useful predictive structure beyond particle-wise features. The result cannot be explained by splitting neighbouring particles between training and test data, because each test fold withholds the complete packing.

Figure 3 complements the numerical tables by showing whether the GNN advantage is consistent rather than produced by only one favourable packing. The GNN recovers more of the searched gain in every fold, although the magnitude varies with the contact topology.

Mean Jaccard overlap with the single best searched ID set was 0.230 for the MLP and 0.254 for the GNN. These modest overlaps do not contradict the thermal improvement. The learning target combines five search restarts, and several different 50-particle assignments can reinforce alternative parallel pathways. Accordingly, thermally evaluated conductivity and recovered gain are the primary criteria; exact agreement with one stochastic teacher

Table 6: Fold-wise ensemble results. Recovery is defined by Eq. (9).

Fold	Test seed	MLP k_{eff}	GNN k_{eff}	MLP rec. [%]	GNN rec. [%]
1	18427	0.12987	0.13969	27.5	42.0
2	29173	0.11791	0.12849	29.5	49.1
3	37811	0.13781	0.14795	36.3	51.3
4	46549	0.12939	0.14353	30.2	55.0
5	55291	0.13445	0.14608	30.7	47.6
6	64109	0.13190	0.13665	41.5	49.0
7	72869	0.13377	0.14499	35.6	53.0
8	81647	0.13540	0.13994	34.3	41.9
9	90439	0.13636	0.15496	37.7	69.0
10	99223	0.12686	0.13665	32.7	49.4

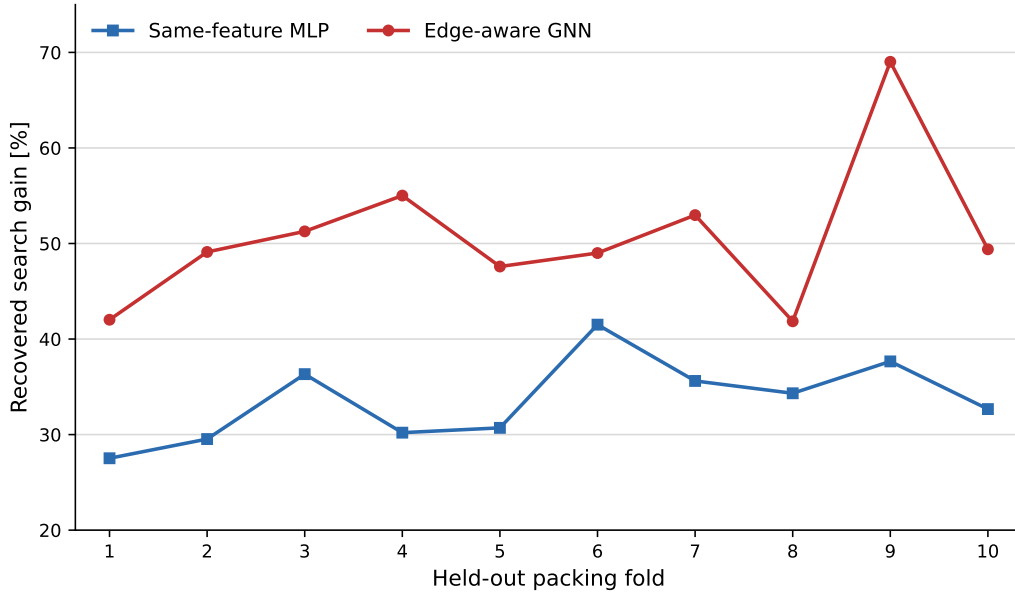


Figure 3: Fraction of the best fixed-budget swap-search gain recovered on each held-out packing. The allocations are produced by three-repeat rank ensembles and evaluated by the unchanged thermal solver.

allocation is secondary.

3.4 Computational cost

The main purpose of the learned models is to avoid testing thousands of material allocations on every new packing. Table 7 therefore compares the work required to select and verify an allocation for one new packing.

Degree ranking requires one thermal-network solution to verify its selected allocation. Heat-path ranking, the MLP and the GNN require two solutions: first, the homogeneous bed is solved to obtain the baseline thermal information, and then the selected allocation is solved to verify its effective conductivity. The neural-network calculation between these two solutions is very fast.

The complete swap-search workflow is substantially more expensive. It includes the random reference calculations, the degree and heat-path baselines, and five search restarts with 3000 proposed swaps each. Across the ten packings, this workflow required 14828.9 ± 45.2 distinct thermal-network solutions per packing. Previously evaluated allocations were retrieved from a cache and were not counted twice.

Table 7: Computational work required to select and verify an allocation for one new packing. Neural scoring time includes the three models used in the ensemble.

Method	Calculation performed	Thermal solves
Degree ranking	Count contacts, select particles and verify the allocation	1
Heat-path ranking	Solve the homogeneous bed, rank particles and verify the allocation	2
MLP ensemble	Solve the homogeneous bed, calculate particle scores and verify the allocation	2
GNN ensemble	Solve the homogeneous bed, calculate graph-based particle scores and verify the allocation	2
Full swap-search workflow	Evaluate random and ranked references and test allocations during five search restarts	14828.9 ± 45.2

The three-model MLP and GNN ensembles required only $(3.43 \pm 0.13) \times 10^{-4}$ s and $(3.96 \pm 0.13) \times 10^{-3}$ s, respectively, to calculate the particle scores on one packing. These CPU times are small compared with the thermal-network solutions.

The models must first be trained using the previously generated search results. Training the three models used in each ensemble required 9.93 ± 1.55 s for the MLP and 30.70 ± 9.19 s

for the GNN. This training is performed once and is not repeated whenever the trained model is applied to another packing.

Consequently, application of the trained GNN to a new packing requires two thermal-network solutions and approximately 4 ms for graph-based particle scoring. This replaces approximately 14829 trial solutions in the complete swap-search workflow, corresponding to about 7400 times fewer thermal solves. The GNN retained $50.7 \pm 7.7\%$ of the conductivity gain found by that search.

3.5 Independent reproduction and conservation

The best ID list from every packing was independently evaluated by the thermal solver. All ten values reproduced the allocation-driver results exactly. Every independent calculation contained exactly 50 high-conductivity particles and the original DEM contact graph. The maximum relative difference between hot- and cold-side heat rates was 3.04×10^{-12} . These checks rule out output rounding, a changed graph or a mismatch in particle IDs as explanations for the observed improvement.

3.6 Mechanism comparison in a representative packing

To examine why the optimized assignment performs better, seed 18427 was analysed in greater detail. Three allocations were compared on exactly the same contact graph: a random realization closest to the random mean, heat-path ranking and the best result from the fixed-budget same-size swap search. The diagnostic quantities in Table 8 describe connections between high-conductivity particles, their contact with the thermal walls, and the heat carried by the resulting network.

The indicators in Table 8 describe three complementary questions. First, how well is the conductive phase connected? The number of *high-high contacts* counts direct contacts between two high- k particles, while the number and size of *high- k clusters* show whether the 50 selected particles remain fragmented or form larger cooperative structures. A *wall-spanning cluster* contains at least one particle touching each thermal wall; the separate hot/cold wall counts give the available heat-entry and heat-exit points.

Second, are those connections useful for the imposed vertical heat flow? The *high-high heat fraction* is the absolute contact heat carried by high-high contacts divided by that carried by all particle-particle contacts. The *heat-weighted vertical alignment* averages $|\Delta z|/d_{ij}$ over high-high contacts using their absolute heat rates as weights; zero denotes a horizontal contact and one a vertical contact. Together, these measures distinguish active vertical pathways from conductive side branches.

Third, what is the quality of the most favourable individual route? Assigning each link the resistance $1/G_{ij}$ gives the reported minimum-resistance hot-to-cold path, and the high- k fraction states how much of that particle path is conductive. This series-path diagnostic is not the total bed resistance, because the complete network contains many parallel routes. The principal performance measure remains k_{eff} , which combines all contacts and all pathways.

Table 8: Mechanism comparison for packing seed 18427. The minimum-path resistance is a descriptive series-path indicator and not the total resistance of the parallel contact network.

Indicator	Random	Heat path	Optimized
k_{eff} [W/(m K)]	0.1112	0.1219	0.1789
High-high contacts	17	27	55
Number of high- k clusters	35	26	5
Largest high- k cluster [particles]	6	5	26
Wall-spanning high- k clusters	0	0	2
High- k particles linked to hot/cold wall	4/2	0/0	7/6
Heat on high-high contacts [%]	4.6	9.1	41.3
Heat-weighted vertical alignment	0.666	0.728	0.804
Minimum-path resistance [K/W]	19894	15884	2892
High- k fraction on minimum path [%]	63.6	55.6	100

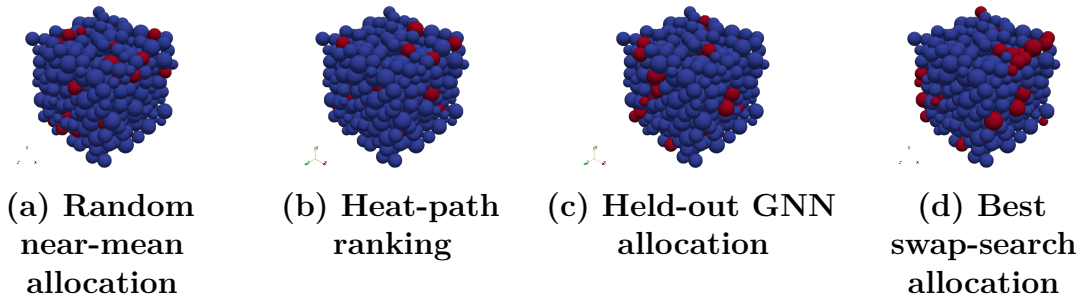


Figure 4: Comparison of four material allocations on packing seed 18427: (a) a random realization closest to the random mean, (b) heat-path ranking, (c) the GNN rank ensemble obtained when this packing was held out, and (d) the best result from the fixed-budget same-size swap search.

The random allocation divided the 50 conductive particles into 35 mostly small clusters, and no all-conductive cluster connected the two walls. Heat-path ranking increased the number and vertical alignment of high-high contacts, but it also produced no wall-spanning

conductive cluster. In contrast, the fixed-budget same-size swap search consolidated the conductive phase into five clusters. The largest contained 26 particles and two clusters connected the hot and cold boundaries.

This structural change also altered where heat travelled. The fraction of the absolute particle-contact heat rate carried by high-high contacts increased from 4.6% for the representative random allocation to 41.3% after optimization. The heat-weighted vertical alignment of those contacts increased from 0.666 to 0.804. Moreover, all nine particles on the calculated minimum-resistance series path were conductive, and its resistance was 85.5% lower than for the random allocation. These observations support a concrete mechanism: the optimizer forms several vertically useful, wall-connected conductive routes and reinforces the pre-existing heat-carrying backbone. The mechanism analysis currently concerns one representative packing and must be repeated for the complete ensemble before it is treated as a general result.

4 Discussion

The main result is simple: the same amount of conductive material can perform very differently depending on its position in the contact network. The weak performance of degree ranking explains why a purely local rule is insufficient. A particle may have many neighbours but still lie in a region with little net vertical heat flow. Heat-path ranking uses the direction and magnitude of the baseline heat flow and therefore performs better. Its consistent 8.3–14.4% improvement is already a useful transparent result.

The much larger gain from the fixed-budget swap search shows that conductive particles must be selected cooperatively. Changing one particle modifies the temperature and heat-flow distribution, which changes the usefulness of other particles. The representative mechanism comparison makes this interpretation more specific: the optimized assignment created two wall-spanning conductive clusters, strongly increased the heat carried by high-high contacts and produced a fully conductive minimum-resistance path. The design problem is therefore global: good assignments build several low-resistance routes between the thermal walls while avoiding isolated conductive clusters and redundant side branches.

The study differs from a conventional conductivity-prediction exercise. The thermal equations remain the evaluator and the goal is to change a discrete material assignment under a strict budget. It also differs from inverse design of freely generated lattices [10]: the contact topology here is mechanically produced by DEM and remains fixed during allocation.

The ML comparison answers why a GNN is useful here. The MLP already performs

better than heat-path ranking because its node features combine size, height, wall access, local degree and baseline thermal information. Nevertheless, it cannot represent how two candidate particles cooperate through their contacts. The GNN improves on the same-feature MLP in every held-out packing and recovers about half of the searched gain. Thus, graph learning is not introduced merely to predict a scalar bed property: it supplies a fast, quota-constrained material allocation that is checked by the governing thermal equations. Recovery below 100% is expected because the GNN performs a single forward ranking, whereas the teacher uses thousands of allocation proposals and repeated full-network solves. The result supports retaining the GNN while also retaining the transparent MLP, heat-path and search references.

Several limitations define the present scope. First, the contact conductance uses a Hertzian radius derived from DEM overlap, so stiffness and packing pressure can affect the predicted magnitude. Second, the bed contains no validated fluid-phase conduction, convection or radiation. Third, only one conductive fraction and one conductivity contrast have been studied. Fourth, the ten independent graphs establish transfer across stochastic realizations of the defined 500-particle system, but not extrapolation to different PSDs, porosities, particle counts or boundary conditions. Finally, the fixed-budget swap search is stochastic; the reported assignments are the best found, not proven global optima. The GNN accordingly learns high-performing searched allocations rather than mathematically certified optima.

5 Conclusions

Ten equal-porosity, equal-PSD DEM packings were used to isolate the influence of conductive-particle placement. With an identical 10/30/10 size-class budget, degree ranking improved effective conductivity only slightly, while heat-path ranking gave a consistent mean improvement of 10.5% over random placement. The best result from five fixed-budget swap-search restarts improved conductivity by 50.8–60.9% in all ten packings. The effect therefore survives changes in random packing topology and cannot be attributed to conductive volume or particle-size bias. Exact independent reproduction and near-machine-precision energy conservation support the numerical reliability of the result. In the representative packing, optimization also produced two wall-spanning conductive clusters and increased the high–high contact heat fraction from 4.6% to 41.3%. This provides initial evidence that the improvement results from connected, vertically useful conductive pathways rather than from high local coordination alone.

The packing-level ML experiment further establishes that the allocation rule is transfer-

able. The same-feature MLP improved k_{eff} by 18.9% over random placement and recovered 33.6% of the best-search gain. The edge-aware GNN improved it by 28.4%, recovered 50.7% of the searched gain and exceeded both the MLP and heat-path rule in every one of the ten held-out packings. The comparison isolates the benefit of message passing because the two learned models use identical node features, training folds, loss, optimizer, stopping criterion, material quotas and thermal evaluator. For the investigated fixed packing specification, graph connectivity therefore contributes useful allocation information that cannot be reduced to independent particle scoring.

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